BCA STUDY POINT

QUESTION BANK

Semester – III Examinatation – 2026

SEC-3

Advanced Database

UNIT I – Database Review, Advanced SQL, Relational Algebra & Calculus

2 Marks

1. What is a Database?
2. What is Three-Schema Architecture?
3. What is an Entity?
4. Differentiate between Tangible and Non-tangible Entity.
5. What is a Weak Entity?
6. What is a Strong Entity?
7. What is a Network Data Model?
8. What is Conceptual Data Modelling?
9. What are Stored, Derived and Complex Attributes?
10. What is a View?
11. What is a Trigger?
12. What is a Stored Procedure?
13. What is an Index?
14. What is Authorization?
15. What is Right Outer Join?
16. What is Relational Calculus?
17. What is Tuple Relational Calculus?
18. What is Domain Relational Calculus?
19. Mention the advantages of E-R Model.

5 Marks

1. Explain Right Outer Join in Relational Algebra with example.
2. Describe the Network Data Model.
3. Compare Weak and Strong Entity Types.
4. Explain the major advantages of E-R Model.
5. Describe Conceptual Data Modelling in brief.
6. Explain Stored, Derived and Complex Attributes.
7. Describe Three-Schema Architecture and explain the role of mapping between the layers.
8. Explain different types of Joins in SQL with examples.
9. Explain Views and their advantages.
10. Explain Stored Procedures and Triggers.
11. Explain Domain Relational Calculus and Tuple Relational Calculus.
12. Explain SQL Integrity Constraints with examples.

UNIT II – Functional Dependencies & Normalization

2 Marks

1. What is Functional Dependency?
2. Define Multivalued Dependency (MVD).
3. What is Trivial Functional Dependency?
4. What is Non-trivial Functional Dependency?
5. What is Functional Dependency Closure?
6. What is Attribute Closure?
7. What is Armstrong's Axioms?
8. What is a Candidate Key?
9. What is a Super Key?
10. What is a Minimal Set of Functional Dependencies?
11. What is a Minimal Cover?
12. What is Database Anomaly?
13. What is Decomposition?

BCA STUDY POINT

14. What is Lossless Decomposition?
15. What is Lossy Decomposition?
16. What is 1NF | 2NF | 3NF | BCNF?
17. What is Denormalization?

5 Marks

1. State Armstrong's Axioms and find the closure of FD set: A→B, AB→C, D→AC, D→E.
2. Compute the closure of R(A, B, C, D, E) with FD set: A→BC, CD→E, B→D, E→A. List the candidate keys.
3. Explain BCNF. Compare 3NF and BCNF.
4. Explain why normalization is required for good relational database design. Explain 2NF with an example.
5. Explain Multivalued Dependency with an example.
6. Explain Armstrong's Axioms with suitable examples.
7. Explain Attribute Closure and Candidate Key with an example.
8. Explain Minimal Cover with an example.
9. Explain Database Anomalies with examples.
10. Explain Lossless and Lossy Decomposition.
11. Explain 1NF, 2NF and 3NF with examples.
12. Explain Denormalization and its advantages.

UNIT III – Transaction, Concurrency Control, Deadlock, Recovery & Security

2 Marks

1. What is a Transaction?
2. What are the different Transaction States?
3. What are ACID Properties?
4. What is a Schedule?
5. What is a Serial Schedule? Non-serial Schedule?
6. What is Conflict Serializability?
7. What is View Serializability?
8. What is a Recoverable Schedule?
9. What is a Cascadeless Schedule?
10. What is a Strict Schedule?
11. What is a Shared Lock? Exclusive Lock?
12. What is Two-Phase Locking (2PL)?
13. What is Timestamp Ordering Protocol?
14. What is Thomas Write Rule?
15. What is Deadlock?
16. Mention the four necessary conditions for Deadlock.
17. What is Deadlock Prevention?
18. What is Deadlock Detection?
19. What is Starvation?
20. What is Database Recovery?
21. What is Rollback?
22. What is Log-based Recovery?
23. What is Checkpoint?
24. What is Cascaded Abort?
25. What is Authorization?
26. What are Grant and Revoke Commands?
27. What is an Audit Trail?

5 Marks

1. State and explain the ACID Properties.
2. Explain Deadlock and the four conditions required for Deadlock to occur.
3. Write a short note on Cascaded Aborts.
4. Explain the purpose of Database Recovery.
5. Explain different Transaction States with a diagram.
6. Explain different types of Schedules with examples.
7. Explain Conflict Serializability using a Precedence Graph.
8. Explain Recoverable, Cascadeless and Strict Schedules.
9. Explain Two-Phase Locking Protocol (2PL) with its two phases.
10. Explain Shared Lock and Exclusive Lock with examples.
11. Explain Timestamp Ordering Protocol.

BCA STUDY POINT

12. Explain Thomas Write Rule.
13. Explain Deadlock Prevention, Detection and Recovery.
14. Explain Wait-for Graph method of Deadlock Detection.
15. Explain Log-based Recovery with an example.
16. Differentiate between Deferred Update and Immediate Update.
17. Explain Shadow Paging.
18. Explain Security Violations and Authorization.
19. Explain Granting Privileges and Roles.
20. Explain Audit Trails.

UNIT IV – Database System Architecture

2 Marks

1. What is Database System Architecture?
2. What is a Centralized Database System?
3. What is a Client-Server Database System?
4. What is a Parallel Database System?
5. What is a Distributed Database System?
6. What is Cloud-based Database System?
7. What is Parallel Processing?
8. What is Distributed Processing?
9. What is Data Replication?
10. What is Fragmentation in Distributed Database?

5 Marks

1. Discuss the advantages and disadvantages of Distributed Database Systems.
2. Explain Centralized Database System Architecture.
3. Explain Client-Server Database Architecture.
4. Explain Parallel Database Systems.
5. Explain Distributed Database Systems.
6. Explain Cloud-based Database Systems.
7. Differentiate between Centralized and Distributed Database Systems.
8. Compare Parallel and Distributed Database Systems.
9. Explain the overview of Parallel and Distributed Storage Systems.

UNIT V – Query Processing, Optimization & Big Data

2 Marks

1. What is Query Processing?
2. What is Query Optimization?
3. What is Query Cost?
4. What is Query Execution Plan?
5. What is Distributed Query Processing?
6. What is a Distributed Transaction?
7. What is Big Data?
8. What is Data Warehousing?
9. What is Data Mining?
10. What is Structured Data?
11. What is Unstructured Data?
12. What is Semi-structured Data?
13. What is Parallel Query Processing?
14. What is Data Warehouse?

5 Marks

1. Give a comparison between Structured and Unstructured Data.
2. Explain the steps involved in Query Processing.
3. Explain different Measures of Query Cost.
4. Explain Query Optimization with an example.
5. Explain Parallel and Distributed Query Processing.
6. Explain Distributed Transactions.
7. Write a short note on Big Data.
8. Explain Data Warehousing and its importance.
9. Explain Data Mining and its applications.
10. Differentiate between Data Warehousing and Data Mining.

BCA STUDY POINT